

The Two-Time Moving-Defect Problem for Dispersing Billiards

A go/no-go audit, dynamic graph currents, and a Fourier route

Working note

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Abstract

We isolate the estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of the scatterers. A tempting bridge from the Demers–Liverani projective cone to forward standard families fails: the moving-face coefficient uses a one-sided trace on a transversal terminal arc, whereas the cone controls stable-leaf integrals. We give a local counterexample to any inference of such a trace bound from the published cone seminorms. We prove a two-sided nuclear/recovery criterion which yields the desired product estimate, but emphasize that it is only a sufficient condition and is probably too strong for a graph delta. The intrinsic problem is instead two-parameter equidistribution of a centered graph current under $T^{-1} \times T$. A targeted literature audit points to an induced generalized-baker/Fourier-decay strategy: encode the collision SRB measure on a two-sided return scheme, prove Fourier decay for the induced slice by non-concentration or L^2 flattening, and lift the estimate through the exponential tower tail and the grazing blocks. An exact third-order calculation at a normal two-cycle between equal circular scatterers gives a nonzero Anosov cocycle, equal to -1 in a determinant-one adapted chart. This supplies a fixed open geometric QNL witness for anchored finite horseshoes; it does not yet give uniform constants for the singular exhaustion or for the weighted moving-face slice. We state the remaining falsifiable estimate and the quantitative margins needed for this program. No unconditional CM2 theorem is claimed here.

1 The exact target

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_X \|f\|_{C^1}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. The same estimate, uniformly for $s^{-1}(P_s - P_0)$, is required before a parameter limit can be exchanged with the two time sums.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_I, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u|_I\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C \varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *The theorem makes clear why a usual functional correlation estimate with error $O(\tau^m + \vartheta^n)$ is insufficient. CM2 requires a centered operator ideal estimate at the event time, not just mixing on either side. Wormell's specialized two-time conditional-mixing conjecture is one way to encode precisely this product structure.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (6) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf_j \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j(W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (11)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (12)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (12) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (13)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (13), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (13) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (13) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b \gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m} \gamma_b(t)) f(T^n F_b \gamma_b(t)) w_b(t) dt. \quad (14)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b \gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (15)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_* \nu_b}(\xi)| \leq C_b (1 + |\xi|)^{-\eta} \quad (16)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (17)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(16) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

The proposition deliberately isolates the missing theorem. Uniform (16) cannot be inferred from one-sided mixing. It must be proved from non-concentration and nonlinearity of the induced inverse branches and the face graph. The L^2 -flattening framework and nonlinear-pushforward theorems of Baker–Khalil–Sahlsten and Baker–Banaji are plausible tools for this step. Wormell proves the corresponding Fourier mechanism for classes of generalized baker maps, but not the present two-time billiard statement.

6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole (m, n) quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

Lemma 6.2 (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (18)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}, \quad (19)$$

with all four exponents positive. Choose $A > \beta_-/\alpha_+$ and $D > \beta_+/\alpha_-$. Suppose in addition that

$$|B_{m,n}| \leq C e^{-\gamma \max\{m, n\}}$$

whenever $D^{-1}m < n < Am$. Then there are $\tau, \vartheta \in (0, 1)$ such that $|B_{m,n}| \leq C' \tau^m \vartheta^n$ for all m, n .

Proof. If $n \geq Am$, (18) is bounded by $e^{-c_+(m+n)}$ for some $c_+ > 0$. If $m \geq Dn$, use (19). The remaining indices lie in the stated diagonal band, where $\max\{m, n\} \geq (m+n)/2$. Taking the minimum of the three positive exponents proves the claim. $\square$

For billiards, (18) is the expected output of signed standard-family coupling after cutting the past-weighted density into homogeneous pieces; the factor $e^{\beta-m}$ records the growth of its regularity. Equation (19) follows by time reversal. Consequently the genuinely new Fourier/QNL theorem is needed only for $m \asymp n$.

This also corrects the naive ambient phase test. The graph $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$ is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable–unstable temporal-distance/holonomy function. A second-derivative QNL condition is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.3 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (20)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (20) follows. On the tangency face, $|((d, 0) - q_j) \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in (24) have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc’s Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom–A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.4 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq Ce^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq Ce^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus the precise new input is a *uniform* singular version of Leclerc’s quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails can then remove the truncation. What remains open is uniform control of the temporal-distance non-concentration constants as the truncation grows.

6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an η_{cut} fraction of its parent’s conditional mass. For an arbitrary exhaustion by finite subshifts, η_{cut} may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations

must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank L . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (21)$$

with η_0 independent of x, L , for at least one QNL-separated child at each scale. If (21) and a uniform UNI constant $\kappa_0 > 0$ hold, Leclerc's tree argument gives a QNL exponent $\Gamma > 0$ independent of L ; only the multiplicative constant is allowed to grow like e^{cL} .

Rühr's pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (21) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

Lemma 6.5 (Uniform interval UNI implies power non-concentration). *Let ν_L be measures on an interval whose densities satisfy $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$, uniformly in L . Let X_L be a family of functions with the following scale-invariant UNI property: for every interval J of length $\sigma < \sigma_0$ and every polynomial P in a fixed finite-dimensional class, there is a subinterval $J' \subset J$ with $|J'| \geq \kappa|J|$ such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where $\kappa > 0$ is independent of J, P, L . Then for every fixed $\alpha > 0$ there are $C, \Gamma > 0$, independent of L , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

Proof. Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval J' . The density bounds imply that at least a fixed fraction $\eta_0 > 0$ of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to $|J|^{1+\alpha}$. The number of generations is $c_\alpha |\log |J|| + O(1)$, so the surviving mass is at most $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C|J|^\Gamma \nu_L(J)$. $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

6.5 A fixed two-branch UNI certificate

Bertozi's 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let h_0, h_1 be two regular inverse return branches on the central magnet, with a common contraction bound $\lambda < 1$, and let τ denote the reduced stable–unstable phase increment

(for the geometric application, $\tau = \log J^u F$ after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (22)$$

Proposition 6.6 (Anchored two-branch certificate). *Suppose the series in (22) converges in C^1 and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

on a central unstable interval I . Then, for all sufficiently large N ,

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.

Proof. C^1 convergence makes $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$ for all large N . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation. $\square$

Corollary 6.7 (First-term domination after acceleration). *Assume $\|Dh_i\| \leq \lambda < 1$, $\|D\tau\|_\infty \leq M$, and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

If

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

then $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$.

Proof. The derivative of the sum of all terms with $k \geq 2$ is bounded by $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1-\lambda)$. Subtract this from the first-term lower bound. $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (20); attach fixed regular connectors to the magnet; and compute Ψ'_∞ (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter Ψ_∞ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order λ for the first-term curvature separation; the remaining infinite temporal-distance tail is only $O(\lambda^2)$. This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not C^2 . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates,

write the map as (F, G) with derivative $\text{diag}(\lambda, \mu)$, where $\lambda > 1$, $\mu = \lambda^{-1}$. The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (23)$$

with all derivatives evaluated at the fixed point.

Proposition 6.8 (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let $\mathcal{R} = T^2$ be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates (s, p) with $p = \sin \varphi$. Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write $S^{-1}\mathcal{R}S = (F, G)$. Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} = F_{yy} = G_{xx} = G_{xy} = G_{yy} &= 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

Proof. Place the circle centres at $(0, 0)$ and $(3, 0)$ and parametrize the first boundary near the facing point by $q(s) = (\cos s, \sin s)$. For outgoing Birkhoff data (s, p) , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at q and a target circle with centre c and radius one, put $w = q - c$, $b = w \cdot v$, and $c_0 = |w|^2 - 1$. The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates (s, p) to $(-s, -p)$ and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by S gives $G_{xyy} = -(7 - 4\sqrt{3})$; substitution in (23) gives -1 . $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength s/R and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and $\mathfrak{A}$ depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the

nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let W^u be a reference unstable leaf and let $H : \Gamma \rightarrow W^u$ be stable holonomy from a central smooth face arc. Write m_Γ for arclength and let w include the physical face velocity, coarea factor, and the smooth branch Jacobians.

Lemma 6.9 (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i) H is a $C^{1+\alpha}$ diffeomorphism onto its image and $0 < c_H \leq JH \leq C_H$;
- (ii) the SRB conditional μ^u on W^u has arclength density between $c_u > 0$ and $C_u < \infty$;
- (iii) $w \in C^\alpha(\Gamma)$ and $\|w\|_\infty \leq C_w$.

Define the signed face measure

$$\nu = H_*(w dm_\Gamma).$$

If, on every interval $J \subset W^u$, the temporal-distance phase X satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing C_1 by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with C_1 multiplied by the sum of their total-variation bounds.

Proof. The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for μ^u compare arclength and μ^u in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces. $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. This does not yet control pieces whose return word or homogeneity rank tends to infinity: their holonomy, density, and amplitude constants must be shown to grow slowly enough for Proposition 6.4 and the radial grazing sum.

7 An induced QNL/Fourier program

The most economical current route is the following.

- (F1) Construct an exponential return scheme on a regular magnet and retain both one-sided codes. In local product coordinates the return map is a countable piecewise-smooth hyperbolic map with collision SRB Gibbs weights. An exact generalized-baker skew product is unnecessary; the appropriate template is the quantitative-nonlinearity/sum-product theorem for nonlinear Axiom-A maps.
- (F2) Lift each complete physical face to finitely or countably many smooth graph slices of the return rectangles. Artificial cuts cancel before the lift, and the two marginals are removed before estimating any branch.
- (F3) Use Lemma 6.2 to reduce to $m \asymp n$. In adapted stable-unstable product coordinates, remove the contracted coordinates and prove uniform nonlinearity/non-concentration for the dominant two-coordinate phase induced by

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m}\gamma_b(t) + \zeta \cdot T^n F_b \gamma_b(t). \quad (24)$$

The associated temporal distance, expressed through stable holonomy, must satisfy UNI and the weighted small-value estimate (27). Exact affine resonance must be excluded; this may require a generic open class of tables rather than all finite-horizon tables. A direct full-product Fourier proof would instead require a uniform finite-type condition through at least fourth order and is not the preferred route.

- (F4) Apply nonlinear pushforward Fourier decay to the stable conditional Gibbs measure; use L^2 flattening when branchwise cancellation is required. Exponential return tails provide the moment bounds, while billiard curvature must provide total nonlinearity or a uniform non-concentration substitute.
- (F5) Lift the return-time estimate to collision time. For the generic assembled trace bound, if block b has physical mass $M_b \lesssim 2^{-b}$ and Fourier constant C_b , the required grazing margin is

$$\sum_{b \geq 0} 2^{-b} C_b < \infty. \quad (25)$$

A polynomial deterioration $C_b \lesssim 2^{\kappa b}$ is acceptable only for $\kappa < 1$. Large grazing expansion can help the oscillatory estimate, but its distortion cost must be tracked in the same inequality. For the Markov-normalized first radial one-face graph, the fixed-atlas flux calculation is sharper: its grouped singular coefficient is $O(2^{-(3-\epsilon_*)b})$, while the weaker $O(2^{-(1-\epsilon_*)b})$ term is a regular two-dimensional interior/Jacobian source. If the oscillatory factorization preserves that separation, the graph-current margin improves to

$$\sum_{b \geq 0} 2^{-(3-\epsilon_*)b} C_b < \infty,$$

allowing $\kappa < 3 - \epsilon_*$. Establishing this preservation for all past words is part of the CM2 proof and cannot be assumed.

- (F6) Establish the same estimates uniformly for the assembled difference quotients

$$s^{-1}(P_s - P_0),$$

and then apply dominated convergence.

The recent symbolic construction of Climenhaga–Day gives a concrete finite-alphabet two-sided coding and product structure for every finite-horizon Sinai billiard map. It is useful as a geometric skeleton, but it constructs the measure of maximal entropy. CM2 concerns the collision SRB/flux measure, so its Gibbs weights, holonomies, and parameter-uniform distortion still have to be built into the induced scheme. This measure mismatch cannot be hidden.

A more faithful use of the smooth Fourier theory is a finite-horseshoe approximation. Truncate the return and homogeneity ranks at L , retain a fixed regular periodic word witnessing nonlinearity, and realize the resulting finite branch core as a smooth area-preserving Axiom–A basic set Ω_L . Put on Ω_L the equilibrium state for the induced unstable-Jacobian potential. Leclerc’s theorem then applies at each fixed L . To pass to the billiard, however, one must prove uniformly

$$\mu\{\text{an orbit segment sees rank} > L\} \leq CNe^{-aL}, \quad (26)$$

$$(\mu_L \otimes \mu_L)\{|\Delta_L| \leq \sigma\} \leq Ce^{cL}\sigma^\Gamma, \quad (27)$$

and the analogous estimate for the weighted face slice. A nonzero periodic-point cocycle in a fixed central branch is a plausible uniform QNL witness, but the published smooth theorem does not state uniformity under this singular exhaustion. Equation (27), rather than the mere existence of a Young tower, is the principal new lemma.

The standard-pair equidistribution theorem of Bálint–Nándori–Szász–Tóth proves that one-dimensional singular initial measures can mix exponentially after a suitable smoothing argument. It supplies the future-time benchmark, not the product estimate: smoothing in the future-stable direction makes the past factor in (14) rougher, and reversing the smoothing exchanges the problem rather than solving it.

Finally, the large-essential-spectrum result of Butterley–Canestrari–Castorrini warns against replacing this dynamic program by a globally stronger Banach norm which controls every discontinuity trace. For piecewise two-dimensional systems such a norm can retain large essential spectrum. This is consistent with the local collar counterexample and favors an orbit-dependent oscillatory estimate.

7.1 Execution audit of the six links

The six links now have the following status.

- (1) *Return coding: available only as an input architecture.* Exponential Young structures and local product rectangles are standard for finite-horizon dispersing billiards. What has not been extracted in the literature is a parameter-uniform two-sided return model carrying the moving face and all its homogeneity labels with the constants needed below.
- (2) *SRB weights: conceptually closed, quantitatively unfinished.* Markov normalization identifies the collision invariant density and the induced Gibbs potential is the unstable-Jacobian potential. The recent finite-alphabet product construction for the measure of maximal entropy cannot replace these weights.
- (3) *Face lift: locally passed.* Artificial faces cancel before the coarea current is formed, the central radial tangency graph has nonzero coarea derivative, and Lemma 6.3 gives an explicit non-affinity certificate. Lemma 6.9 transfers central-block QNL to its physical weighted face measure. A global bi-return partition of the face remains to be written.

- (4) *Quantitative nonlinearity: local witness closed, uniform transfer unfinished.* Proposition 6.8 gives an exact nonzero Anosov cocycle at a fixed normal two-cycle and hence a geometric QNL witness which survives anchored truncation. The remaining new statement is uniform transfer from the now-closed central block to the countable high-return/high-homogeneity pieces.
- (5) *Fourier conversion: conditionally closed.* Proposition 6.1 gives CM2 once QNL yields uniform Fourier decay. The frequency scale is the maximum of the two hyperbolic scales, which still gives a product rate with half the Fourier exponent.
- (6) *Tower and grazing lift: a quantitative margin is identified.* Proposition 6.4 handles exponential return tails. The generic graph-current block margin is (25); the first radial one-face calculation potentially supplies the stronger exponent $3 - \epsilon_*$. Uniform difference-quotient QNL is still needed for parameter differentiation.

Accordingly the original six-step program does not yet prove CM2. The local nonlinearity is no longer conjectural: it has the exact witness (23). The positive theorem is now reduced to a singular transfer statement: use that fixed witness to obtain a uniform temporal-distance non-concentration estimate for the double-centered radial face on anchored return/homogeneity truncations, with constants growing slowly enough for Proposition 6.4 and the grazing margin.

7.2 A quantitative smoothing test

There is one falsifiable way to obtain (13). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (28)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (29)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad qquad \Lambda_+ e^{-a\alpha} < 1. \quad (30)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (29) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (29). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 7.1 (Uniform difference-quotient closure). *Suppose (11)–(12) hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$, and each block converges weakly to $K_{V,b}$. Then the parameter difference quotient of the prescribed susceptibility is dominated by a summable $C\tau^m \vartheta^n$ majorant and converges term by term to the double series generated by K_V .*

Proof. Apply Theorem 3.2 uniformly in s and sum over b, m, n . Equation (12) and the two geometric series give an integrable majorant. Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

8 Next falsifiable tests

The first calculation should use one central physical face and a finite strip cutoff. After complete assembly and double centering, it should prove or disprove the local estimate

$$\left| \int_I g(T^{-m}\gamma(t))f(T^n F\gamma(t))w(t) dt \right| \leq C\tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (31)$$

The proof attempt must analyze the phases (24). If a resonant branch admits covectors for which both the first and second phase derivatives remain small on positive slice mass, the Fourier route fails for that table. If the non-concentration estimate holds uniformly, CM2 has a concrete proof route and the dyadic blocks should then be added using (25).

In parallel, one may still test the stronger mixed-cylinder condition (13). Success would imply CM2 immediately, but failure would only show that the graph current is non-nuclear; it would not refute (31).

The current status is therefore precise: future recovery is technically available, complete centering is algebraic, the prescribed source is already independent and is not a coboundary, and a fixed circular normal cycle gives an exact nonzero QNL witness. The unresolved theorem is the uniform passage from that finite smooth witness to dynamic two-sided equidistribution of the weighted complete terminal graph current under singular exhaustion. Existing projective cones, coupling of standard families, and anisotropic current spaces do not prove it. The best falsifiable route found in the present audit is the anchored induced/Fourier program above.

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